



SERVICE BULLETIN



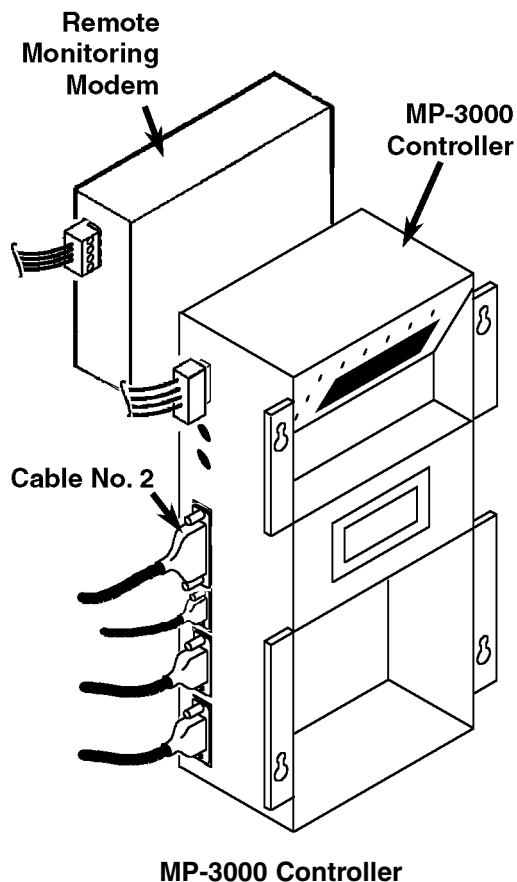
SUBJECT: Manual Emergency Mode Operation or Replacement of MP-3000 Controllers

BULLETIN NO. C 041

DATE: December 4, 2000

This is to remind all personnel that they **MUST ALWAYS** disconnect the unit power cord from the power supply before attempting to disconnect or replace a MP-3000 controller.

WARNING: The unit will automatically start and operate if 460/380V power is present at the main relay board and Cable No. 2 is disconnected from the controller. To prevent personal injury from rotating machinery or dangerous electrical shock from high voltage controls, disconnect the supply power to the unit. Then proceed to replace the controller or prepare the unit for manual emergency mode operation.



Manual Emergency Mode Operation

In the event of an emergency situation where a failure of the controller occurs, a manual emergency mode function can be used to operate the unit. However, the unit must be manually cycled ON and OFF using the unit 460/380V main circuit breaker. This is because manual control disconnects both the controller and Unit On/Off switch from the main relay board.

NOTE: On CSR units, both the 460/380V main circuit breaker and the Unit On/Off switch must be used to cycle the unit ON and OFF. The Unit On/Off switch must be ON to operate the scroll compressor.

Manual control offers a selection of six operating positions. Refer to the Manual Control decal in the unit control box or the unit Maintenance Manual to set the J501 Jumper for the desired unit operating mode.

NOTE: CRR units and CSR units use different J501 Jumper positions to provide cooling on emergency mode operation.

The unit must be cycled ON and OFF manually to maintain the desired temperature. Use the 460/380V main circuit breaker to start and stop the unit. Monitor container temperature with an external thermometer.

NOTE: The unit cooling capacity on Chill loads can be reduced by almost closing the suction service valve when Cool 1 is selected. If the compressor overheats, select Cool 2.

1. To select Manual Control, turn the unit On/Off switch to OFF.
2. Turn the unit 460/380V main circuit breaker OFF. Then disconnect the unit power cord from the power supply.

WARNING: The unit will automatically start and operate if 460/380V power is present at the main relay board when the controller is disconnected. To prevent personal injury from rotating machinery or dangerous electrical shock from high voltage controls, disconnect the supply power to the unit before preparing the unit for manual emergency mode operation.

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3. Disconnect cable no. 2 from the controller and main relay board (see electrical schematic). The main relay board will now control the unit based on the manual control setting.

NOTE: You MUST check 2-pin plug location on J501 connections of main relay board to ensure correct unit operation.

4. If necessary, remove 2-pin plug from J501 (see decal on main relay board) and re-locate based on the unit operating mode required. Refer to the Manual Control decal in the unit control box.
5. Connect the unit power cord to the proper power supply.
6. Start the unit by turning the unit 460/380V main circuit breaker ON.

NOTE: On CSR units, both the 460/380V main circuit breaker and the Unit On/Off switch must be used to cycle the unit ON and OFF. The Unit On/Off switch must be ON to operate the scroll compressor.

7. Check for correct rotation of condenser fan. Condenser air should be blowing out from the center of the grille. If the fan is running backwards, the power supply phase must be changed.

Reversing Power Phase on CRR Units

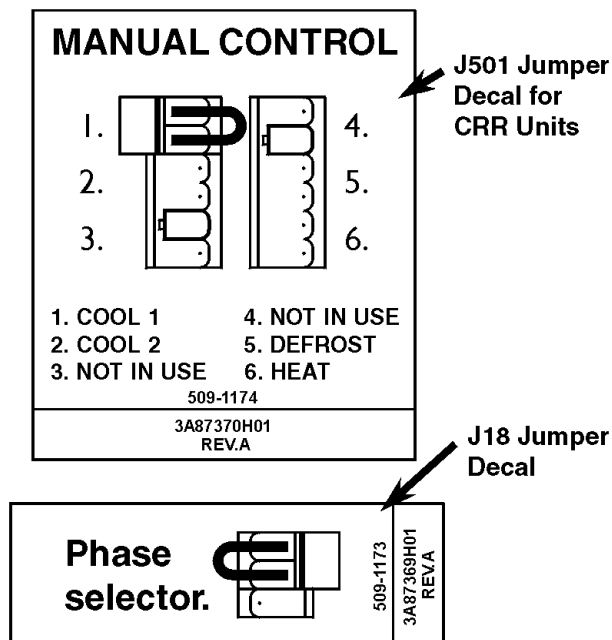
On CRR units, use Jumper J18 to reverse the power phase.

1. Turn the unit 460/380V main circuit breaker OFF.
2. Disconnect unit power cord from power supply.
3. Relocate the phase selector terminal plug from J18 (see decal on main relay board). Relocate from A to B (B to A) as required.
4. Connect unit power cord to proper power supply.
5. Start the unit again by turning the unit 460/380V main circuit breaker ON. Check condenser airflow again to confirm correct fan rotation.

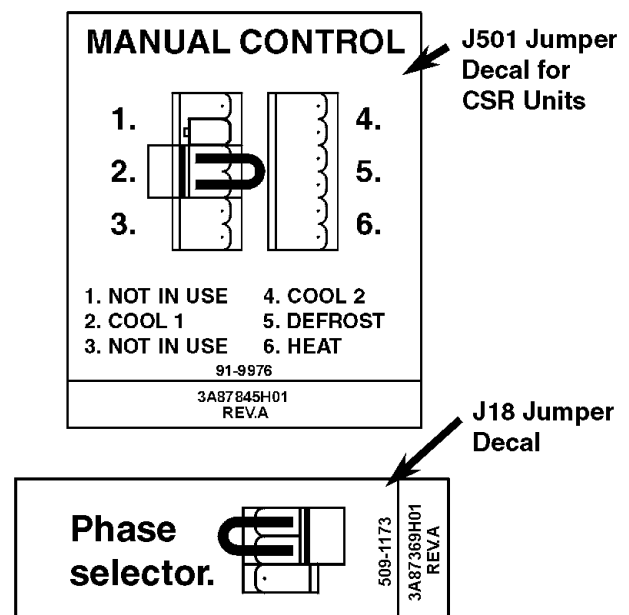
Reversing Power Phase on CSR Units

On CSR units, use incoming power cable leads to reverse the power phase. This is recommended on CSR units because the Jumper J18 does not reverse power to the scroll compressor. Using the incoming power cable leads therefore protects against the possibility that the compressor will be out of phase with the condenser and evaporator fans when the unit is plugged into new power supply.

1. Turn the unit 460/380V main circuit breaker OFF.
2. Disconnect unit power cord from power supply.
3. Relocate the position of the white and black incoming power cord leads at the 460/380V main circuit breaker.
4. Connect unit power cord to proper power supply.
5. Start the unit again by turning the unit 460/380V main circuit breaker ON and the unit On/Off switch ON. Check condenser airflow again to confirm correct fan rotation.



CRR Decals for Manual Emergency Control



CSR Decals for Manual Emergency Control